

*An Anatomy of Monopsony:
Search Frictions, Amenities and Bargaining in Concentrated Markets*

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Introduction

Modern monopsony literature based on three paradigms

- i. Worker-firm specific preference heterogeneity Card Cardoso Heining Kline (2018)
- ii. Search frictions Burdett Mortensen (1998)
- iii. Firm granularity Berger Herkenhoff Mongey (2022)
Jarosch Nimczik Sorkin, (2019)

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- *Which mechanisms account for the wage markdown ($\equiv w / mrpl$)?*

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Approach

1. Propose a model that captures these three paradigms
2. Estimate the model on Norwegian data
3. Quantify the role of each source in average markdowns

Introduction

Theory: Granular labor market in GE with worker-firm specific preferences

- Cahuc, Postel-Vinay, Robin (2006) + Worker-firm specific amenities + M optimizing firms
- Relax neoclassical assumption of zero price-discrimination within firm (e.g. BHM 2022)
- Incorporate:
 - i. Neoclassical sources of markdowns (first-degree price discrimination)
 - ii. On-the-job search
 - iii. Contact rates in equilibrium → welfare and policy

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Estimation:

- (1) Cluster occupations into local labor markets in *Norway* based on similarity of job flows
- (2) Empirically show negative covariance of HHI with wage, job flows, dispersion of wage,
- (3) Model targets/replicates this

Environment

- **Markets** - Many labor markets indexed j . Independent. Single final good.
- **Agents** - Measure $\overline{N}_j$ identical workers. Either $\in \{u, e\}$. M_j firms $k = 1, \dots, M_j$

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Within market j ...
- **Production** - Firm k produces $y_k = z_k n_k$ with n_k workers. At home, produce b .
- **Vacancies** - Firms post v_k vacancies each period with flow cost $c(v_k)$
- **Search** - Fraction $\{\zeta_u, \zeta_e\}$ of $\{u, e\}$ apply for jobs, x_k reach firm k , $A m(x_k, v_k)$ meet
- **Preferences** - Before matching, **worker i draws amenity** $\varepsilon_{ikt} \sim F(\varepsilon)$, constant during match and observed

$$u_{ikt} = w_{ikt} + \varepsilon_{ikt} \quad , \quad Utility_i = \mathbb{E} \left[\sum_{t=0}^{\infty} \beta^t u_{ikt} \right]$$

- **Bargaining** - As in Cahuc-PVR: Bertrand + Nash, worker bargaining power $\theta \in [0, 1]$.

Search and matching

Timing: (i) **apply**, (ii) **meet**, (iii) **draw preference shock/non-wage amenity**, (iv) **match**

i. Fraction $\{\tilde{\zeta}_u, \tilde{\zeta}_e\}$ of $\{u, e\}$ **randomly apply** for jobs

- Unemployed apply to all $k \in \{1, \dots, M\}$ firms
- Employed at firm k apply to firms $k' \neq k$ (partially directed, k removed)

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Contact rates λ_{ik} for workers, λ_{ik}^f for firms, (*first subscript= origin, second=destination*).

Accounting: $\lambda_{ik} n_i = \lambda_{ik}^f v_k$

Bargaining protocol

Let U denote value of unemployment and W denote **current** promised value at (k, ε)

Cahuc, Postel-Vinay, Robin: 3 cases if worker at firm (k, ε) meets firm (k', ε')

(1) $S(k, \varepsilon) > W - U > S(k', \varepsilon')$: greater surplus and promised value at **current firm**

- ▶ Do not move
- ▶ Do not negotiate

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(2) $S(k, \varepsilon) > S(k', \varepsilon') > W - U$: greater surplus but lower promised value at current firm

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- ▶ **renegotiate** $W = S(k', \varepsilon') + U$

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(3) $S(k', \varepsilon') > S(k, \varepsilon)$: new firm offers greater surplus

- ▶ **move**
- ▶ **bargain** $W = \theta(S(k', \varepsilon') - S(k, \varepsilon)) + S(k, \varepsilon) + U$

Surplus (worker value + firm value - outside option)

Bellman equation

$$S(k, \varepsilon) = (z_k + \varepsilon - b) + \beta \left[S(k, \varepsilon) - \delta S(k, \varepsilon) + \underbrace{\theta \int \sum_{k' \neq k} \lambda_{kk'} \max \{ S(k', \varepsilon') - S(k, \varepsilon), 0 \} dF(\varepsilon')}_{\text{Increase in surplus when moving to higher } (k', \varepsilon')} \right. \\ \left. - \underbrace{\theta \int \sum_{k'} \lambda_{uk'} \max \{ S(k', \varepsilon'), 0 \} dF(\varepsilon')}_{\text{Opportunity cost of employment}} \right]$$

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Worker promised value: CPVR determines worker's promised value

Follow [Lise, Robin 2017](#), write worker's promised value $W(k, \varepsilon, \sigma)$ as share σ of surplus

$$\sigma := \frac{\text{Worker surplus}}{\text{Total surplus}} = \frac{W(k, \varepsilon, \sigma) - U}{S(k, \varepsilon)} \implies S(k, \varepsilon), \sigma \rightarrow \underbrace{w(k, \varepsilon, \sigma)}_{\text{Wage}}$$

Wages

Promised values delivered by fixed wage until outside offer ($\sigma :=$ worker's surplus share)

$$w(k, \varepsilon, \sigma) = \overbrace{\sigma z_k}^{\text{output share}} - \overbrace{(1 - \sigma)\varepsilon}^{\text{amenity discount}}$$

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 \text{Backloaded quits} &- (1 - \sigma) \beta \int \sum_{k' \neq k}^M \lambda_{kk'} \mathbf{1}_{[S' > S]} \left[S(k, \varepsilon) + \theta \left(S(k', \varepsilon') - S(k, \varepsilon) \right) \right] dF(\varepsilon')
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 \end{aligned}$$

- Forces: (i) Search: λ 's, (ii) Bargaining: θ , (iii) Amenities: ε 's, (iv) Concentration: M
- Concentration limits surplus shares
- Inherent duopsony in wage setting

Firms

Assumptions

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Resulting firm objective function

$$\begin{aligned} V_k = \max_{v_k} & -c(v_k) + \underbrace{v_k \lambda_{uk}^f (1 - \theta) \int \max \{S(k, \varepsilon), 0\} dF(\varepsilon)}_{\text{Hire from unemployment}} \\ & + \underbrace{v_k \sum_{\varepsilon', k' \neq k} \lambda_{k'k}^f \left(\frac{n(k', \varepsilon')}{n(k')} \right) (1 - \theta) \int \max \{S(k, \varepsilon) - S(k', \varepsilon'), 0\} dF(\varepsilon)}_{\text{Hire from competitors } k'} \end{aligned}$$

QUANTITATIVE

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$$\underbrace{\frac{\text{Avg. self-flow rate of } K \text{ clusters}}{\text{Std. dev. self-flow rate of } K \text{ clusters}}}_{\text{Reward fit}} \times \underbrace{\left[1 - \text{Herfindahl of size of } K \text{ clusters} \right]}_{\text{Penalize overfitting}}$$

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Discussion of markets

Commuting Zone: *Oslo*

Dental ABC

Law XYZ

Dentists

$j = \text{Dentists in Oslo}, M_j = 1$

Lawyers

$j' = \text{Lawyers in Oslo}, M_{j'} = 1$

Secretaries

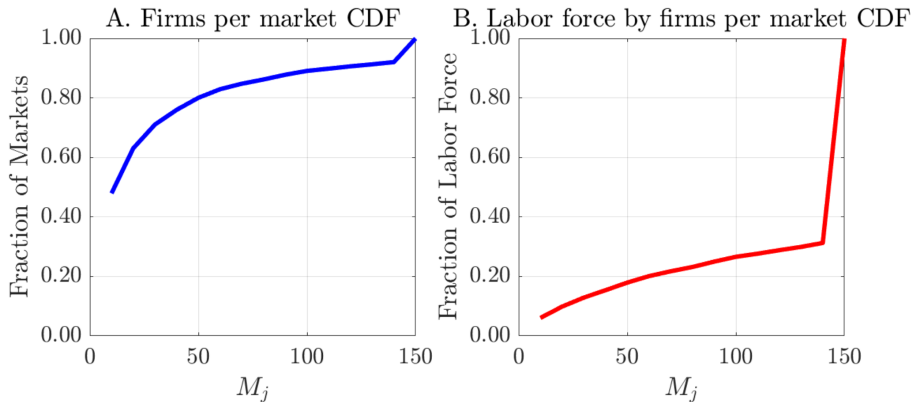
Secretaries

$j'' = \text{Secretaries in Oslo}, M_{j''} = 2$

Markets

Important feature of Norway: Most workers are in non-concentrated markets

- 70% of work force is in markets with at least 150 firms per market



Calibration

- Calibrate monthly model to average of aggregate moments from 2006-2016
- Take M_j pdf straight from data ≈ 73 firms per market, impose $\bar{N}_j = aM_j$
- Cobb-Douglas matching $\alpha = 0.5$, $\xi_u = 1$. Isoelastic cost $c(v) = \frac{a^\gamma}{1+\gamma} v^{1+\gamma}$
- Amenities uniform $\varepsilon \sim [0, \bar{\varepsilon}]$

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- Estimate remaining 8 parameters using SMM

Parameter		Value	Moment	Model	Data
Match efficiency	A	0.18	Unemployment rate	0.03	0.04
OJS intensity	ξ_e	0.32	Aggregate EE rate	0.01	0.01
Bargaining power	θ	0.18	Average log wage growth	0.01	0.01
Home production	b	0.85	Replacement rate	0.86	0.66
Upper bound amenity	$\bar{\varepsilon}$	0.76	Fraction of EE moves down job ladder (poach index)	0.21	0.15
Std Dev Productivity	σ_z	0.14	Standard deviation of log wages	0.69	0.63
Vacancy cost elasticity	γ	1.16	Employment HHI	0.09	0.08
Cross-market regression coefficients			Log average wage vs. HHI	-0.09	-0.09
			Wage dispersion vs. HHI	-0.18	-0.18

Concentration and wages

Challenge: No valid instrument for concentration

- ▶ Local effects of national exit → alters rungs on job ladder
- ▶ **Our approach:** provide benchmark set of covariances that any model of concentration and labor markets should replicate

Discipline model with covariance of HHIs and wages using two sources of variation:

- ▶ **W/in year-occupation**, x-region variation (Dentists in Oslo v. Bergen)
 - ▶ Control for sorting with observables
- ▶ **W/in occupation-region**, x-time variation (Dentists in Bergen today/tomorrow)
 - ▶ Unlikely to reflect sorting
- ▶ **Validate model against covariance of HHIs and job flows**

Concentration and wages

Specifications. Occ o , Region r , Month t . Two regressions w/ $FE \in \{\text{Occ-Year}, \text{Occ-Region}\}$:

$$Y_{ort} = FE + \beta HHI_{m(o,r)t} + \gamma X_{ort} + \epsilon_{ort}$$

Controls. X_{ort} includes age, gender, education, month, lagged firms-per-worker & LF growth

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Dependent - Y_{ort}	Model	Occ-Year
Log wage (target)	-0.0915	-0.0104
Std. dev. log wage (target)	-0.1799	-0.125***
EE rate	-0.0027	-0.00214***
UE rate	-0.0103	-0.0330***

Interpretation: 1 SD higher HHI (SD=0.27) is associated with 0.27 p.p. wage decrease

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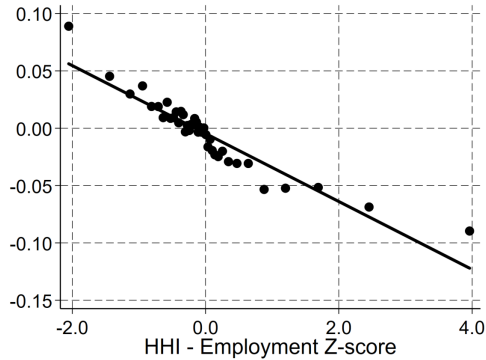
Dependent - Y_{ort}	Model	Occ-Year	Occ-Region
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Std. dev. log wage (target)	-0.1799	-0.125***	-0.230***
EE rate	-0.0027	-0.00214***	-0.00475***
UE rate	-0.0103	-0.0330***	-0.0284***

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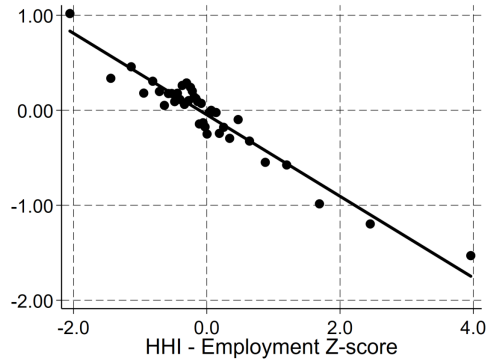
Summary: Model does well at replicating negative covariances of HHI with job flows

Visualizing the data

A. E-to-E Rates (%)



B. U-to-E Rates (%)



Counterfactuals

Baseline wage split into four components:

$$w(\sigma, k, \varepsilon) =$$

1. Output share (91.4%) σz_k

2. Opportunity cost (42.0%) $+ (1 - \sigma) \left(b + \beta \theta \sum_{k'}^M \lambda_{uk'} \int \max \{ S(k', \varepsilon'), 0 \} dF(\varepsilon') \right)$

3. Amenity discount (-18.8%) $-(1 - \sigma)\varepsilon$

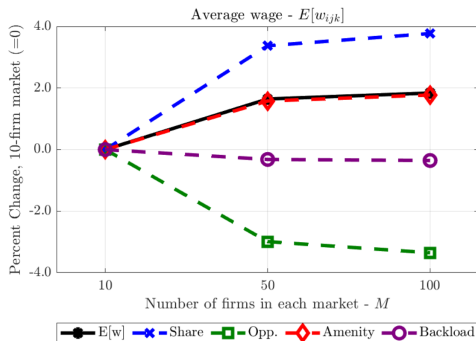
4. Quit / Promotion discount (-14.6%) $- (1 - \sigma) \beta \int \sum_{k' \neq k}^M \lambda_{kk'} \mathbf{1}_{[S' > S]} \left[S(k, \varepsilon) + \theta (S(k', \varepsilon') - S(k, \varepsilon)) \right] dF(\varepsilon')$
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Counterfactuals - Wages

- Isolate M (*pure concentration*): (i) Fix $[z_1, \dots, z_{10}]$, duplicate 10x, (ii) Arrange into Ten $\times$ 10-firm markets, Two $\times$ 50-firm markets, One $\times$ 100-firm market
- (1) Same z 's in each mkt., removes mechanical granularity/ladder height and (2) $M/\bar{N}$ constant

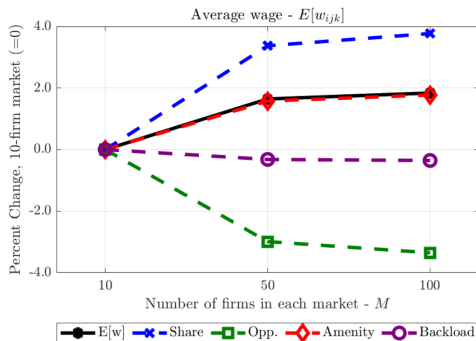
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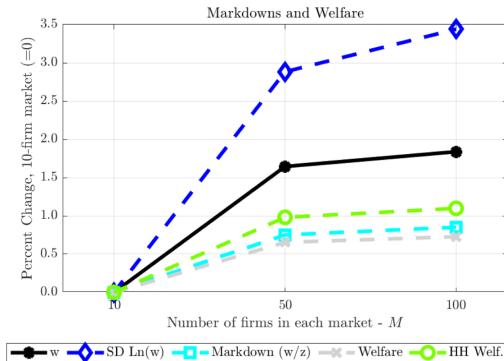
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- (1) Same z 's in each mkt., removes mechanical granularity/ladder height and (2) $M/\bar{N}$ constant



- Surplus share $\uparrow \sigma$ +3pp via comp., $\uparrow \sigma z$ +4pp as flow rates $\uparrow$ to high z 's $\implies \downarrow (1 - \sigma)b$, $(1 - \sigma)\epsilon$ less negative
- Backloading increases, putting downward pressure on wages

Counterfactuals - Welfare and Inequality

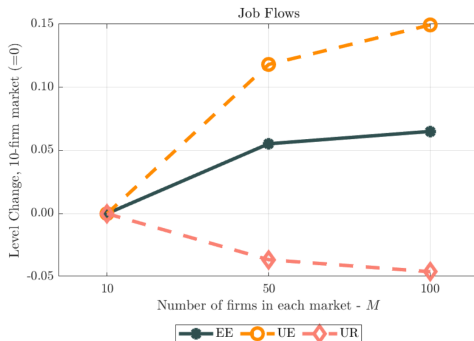
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- Welfare $\uparrow$ 0.75%, Markdowns ($:= w/z$) improve (workers capture more of MRPL)
- Wage inequality increases as markets become more competitive

Counterfactuals - Contact rates

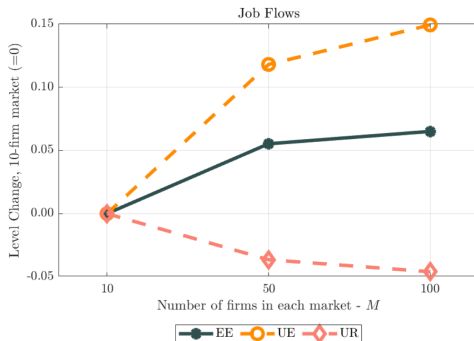
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- Next: Competitive limits of agg. economy to measure importance of Concentration vs. Neoclassical channels

Counterfactuals - Competitive limits in Norway

(1) $M_j \times 2$ [scale to remove Δ s in $M/\bar{N}$, $\{z\}$], (2) $\lambda_{ii} > 0$, (3) $\sigma_\epsilon = 0$, (4) $\xi_e \rightarrow 1$, $A \rightarrow \infty$

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Variable	Baseline	A. Duplicate z , $2\{M_j, \bar{N}_j\}$	B. Meet yourself	C. One ϵ	D. No Search $\zeta_e = 1$, $A = \infty$
HHI_n	0.09	0.05	0.12	0.28	0.75
Markdown (w/z) (Emp. weighted)	0.79	0.80	0.80	0.86	0.93
EE Rate (%)	0.55	0.56	0.62	0.63	1.17
Labor Productivity vs. Base (%)	-	0.42	0.92	7.63	12.71
Welfare vs. Base (%)	-	0.25	0.57	6.62	14.71
Average wage vs. Base (%)	-	0.68	1.40	16.80	30.90
Wage dispersion vs. Base (%)	-	0.81	2.10	-9.00	-91.71

(A.) **Pure concentration.** Duplicate $\{z\}$ vector in each market, double M_j and $\bar{N}_j$
 - Wages +0.70%, Welfare +0.25%, Inequality +0.81%, Markdowns narrow 1 ppt

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- Competing with continuum of own vacancies, Markdowns narrow 1 ppt

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(D.) **Less search frictions.** Increase HHI , big firms siphon workers, Markdowns narrow 14 ppt

Counterfactuals - Competitive limits in Norway

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Variable	A. One ε	B. One ε $+\theta = 0.5$	C. One ε $+\theta = 0.9$	D. One ε $+\text{Meet next ranked}$
HHI_n	0.28	0.54	0.72	0.84
Markdown (w/z) (Emp. weighted)	0.86	0.97	1.00	0.97
EE rate (%)	0.63	0.64	0.67	0.48
Labor productivity v. One ε (%)	-	3.75	5.08	4.9
Welfare v. One ε (%)	-	5.48	7.34	7.2
Average wage v. One ε (%)	-	15.45	20.3	18.3
Wage dispersion v. One ε (%)	-	-87.2	-93.8	-95.2

$\theta \rightarrow 1$ mechanically eliminates markdowns, leaves search frictions untouched

- Suggests potential important role for strategic interactions

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HHI_n	0.28	0.54	0.72	0.84
Markdown (w/z) (Emp. weighted)	0.86	0.97	1.00	0.97
EE rate (%)	0.63	0.64	0.67	0.48
Labor productivity v. One ε (%)	-	3.75	5.08	4.9
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Multiple parties at bargaining table

- Meet 'next lower ranked' firm (meet rank $K + 1$ whenever meet rank K , where 1=best)
- (allocations unaffected in PE, only outside option)
- Reduces markdowns 11pp— large role for strategic interactions

Conclusion

Theory contribution: $CPVR + M + \varepsilon$

- GE theory w/ three sources of monopsony (Concentration/Search/Neoclassical)
- By determining quantities and prices, useful for policy
- Tractable model, reproduces key negative cov. b/w HHI's, job flows and wages

Empirical contribution:

- New clustering algorithm, establish benchmark set of HHI covariances

Quantitative contribution: decompose markdowns...

- ▶ 1/3 due to neoclassical monopsony
- ▶ 1/2 due to duopsonistic wage bargaining
- ▶ 2/3 due to search frictions

THANKS!